Module 1 : Linear Algebra (theory of matrices)

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* Characteristic eq^n

Consider square matrix A then its characteristic eq^n is $|A - \lambda I| = 0$, where λ is called Eigen value and I is Identity matrix.

ex., if $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ then its characteristic eq^n can be written as $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) - 8 = 0$$

$$3 - \lambda - 3\lambda + \lambda^2 - 8 = 0$$

$$\lambda^2 - 4\lambda - 5 =$$

$$\lambda = 5 \text{ or } \lambda = -1$$

$\therefore$ Eigen values are 5 & -1

* Eigen value and eigen vector.

~~The root~~

- Eigen Value:

The root or solution of characteristic equation is called Eigen value. Eigen value is also called latent value or proper value or characteristic value.

- Eigen vector.

Eigen vector depends on eigen value. Eigen vector is a column matrix (single column). Eigen vector is obtained from following eq^n $[A - \lambda I]X = 0$ where, A is a square matrix and λ is eigen value, I is Identity matrix and X is eigen vector.

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Ex. 1] Find the eigen value of the matrix $\begin{bmatrix} 3 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

→ The characteristic eqn is $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 3-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - [3+3+3]\lambda^2 + [(3)(3)+(3)(3)+(3)(3) - (-1)(-1) - (1)(1) - (-1)(-1)]\lambda - 20 = 0$$

$$\lambda^3 - 9\lambda^2 + 24\lambda - 20 = 0$$

$$\lambda = 5, 2, 2$$

Ex. 2] Find eigen value and eigen vector of the matrix $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$

→ The characteristic eqn is $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) - 8 = 0$$

$$3 - \lambda - 3\lambda + \lambda^2 - 8 = 0$$

$$\therefore \lambda^2 - 4\lambda - 5 = 0$$

∴ This is required characteristic eqn

$$\lambda = 5 \quad \text{or} \quad \lambda = -1$$

Now,

eigen vector for eigen value $\lambda = 5$

$$[A - \lambda I]X = 0$$

$$\begin{bmatrix} 1-5 & 2 \\ 4 & 3-5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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$R_2 \rightarrow R_2 + R_1$

$$\begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-4x_1 + 2x_2 = 0$$

$$2x_2 = 4x_1$$

$$x_2 = 2x_1$$

Let,

$x_1 = t$

$x_2 = 2t$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} t \\ 2t \end{bmatrix} = t \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

put $t = 1$

$$X = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$\therefore$ required eigen vector for eigen value $\lambda = 5$ is

$$X = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Now,

eigen vector for eigen value $\lambda = -1$

$$[A - \lambda I] X = 0$$

$$\begin{bmatrix} 1-(-1) & 2 \\ 4 & 3-(-1) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$R_2 \longrightarrow R_2 - 2R_1$

$$\begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x_1 + 2x_2 = 0$$

$$2x_2 = -2x_1$$

$$x_2 = -x_1$$

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Let,

$x_1 = t$

$x_2 = -t$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} t \\ -t \end{bmatrix} = t\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

put $t = 1$

$$X = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$\therefore$ required eigen vector for eigen valued $\lambda = -1$ is $X = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

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Ex. 2] If $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ find $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$

$\rightarrow$ The characteristic eqn is $|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - [2+1+2]\lambda^2 + [(2)(1) + (1)(2) + (2)(2) - (0)(1) - (1)(1) - (1)(0)]\lambda - 3 = 0$$

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0 \quad \text{——— (1)}$$

$\therefore$ This is required characteristic eqn

put $A = \lambda$

$$\frac{\lambda^8}{\lambda^3} = \lambda^5$$

$$\begin{array}{r|l} & \lambda^5 + \lambda \\ \hline \lambda^3 - 5\lambda^2 + 7\lambda - 3 & \lambda^8 - 5\lambda^7 + 7\lambda^6 - 3\lambda^5 + \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1 \\ & -\lambda^8 \mp 5\lambda^7 \pm 7\lambda^6 \mp 3\lambda^5 \\ \hline & \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1 \\ & -\lambda^4 \mp 5\lambda^3 \pm 7\lambda^2 \mp 3\lambda \\ \hline & \lambda^2 + \lambda + 1 \end{array}$$

$$\frac{\lambda^4}{\lambda^3} = \lambda$$

$$\lambda^8 - 5\lambda^7 + 7\lambda^6 - 3\lambda^5 + \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1 = (\lambda^3 - 5\lambda^2 + 7\lambda - 3)(\lambda^5 + \lambda) + (\lambda^2 + \lambda + 1)$$

from eqn (1)

$$\lambda^8 - 5\lambda^7 + 7\lambda^6 - 3\lambda^5 + \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1 = \lambda^2 + \lambda + 1$$

then required matrix is

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 8A + I = A^2 + A + I$$

$$A^2 + A + I = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^2 + A + I = \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

Tute

Ex.3] Find $2A^4 - 5A^3 - 7A + 6I$ where $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$

→

The characteristic eqⁿ is $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & 2-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(2-\lambda) - 4 = 0$$

$$2 - \lambda - 2\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 3\lambda - 2 = 0 \quad \text{——— (1)}$$

∴ This is required characteristic eqⁿ

Put $A = \lambda$

$$\begin{array}{r|l} & 2\lambda^2 + \lambda + 7 \\ \hline \lambda^2 - 3\lambda - 2 & 2\lambda^4 - 5\lambda^3 - 7\lambda + 6 \\ & -2\lambda^4 \mp 6\lambda^3 \mp 4\lambda^2 \\ \hline & \lambda^3 + 4\lambda^2 - 7\lambda + 6 \\ & -\lambda^3 \mp 3\lambda^2 \mp 2\lambda \\ \hline & 7\lambda^2 - 5\lambda + 6 \\ & -7\lambda^2 \mp 21\lambda \mp 14 \\ \hline & 16\lambda + 20 \end{array}$$

$\frac{\lambda^4}{\lambda^2} = \lambda^2$

$\frac{\lambda^3}{\lambda^2} = \lambda$

$$2\lambda^4 - 5\lambda^3 - 7\lambda + 6 = (\lambda^2 - 3\lambda - 2)(2\lambda^2 + \lambda + 7) + (16\lambda + 20)$$

from eqⁿ (1)

$$2\lambda^4 - 5\lambda^3 - 7\lambda + 6 = 16\lambda + 20$$

Then required matrix is

$$2A^4 - 5A^3 - 7A + 6I = 16A + 20I$$

$$16A + 20I = \begin{bmatrix} 16 & 32 \\ 32 & 32 \end{bmatrix} + \begin{bmatrix} 20 & 0 \\ 0 & 20 \end{bmatrix} = \begin{bmatrix} 36 & 32 \\ 32 & 52 \end{bmatrix}$$

$$2A^4 - 5A^3 - 7A + 6I = \begin{bmatrix} 36 & 32 \\ 32 & 52 \end{bmatrix}$$

* Similarity of matrices.

If A & B are two square matrices of same order then matrix A is said to similar to matrix B, if there exists non-singular matrix N such that $A = MBM^{-1}$ or $M^{-1}AM = B$.

The matrix M is called modal matrix

+ Diagonalizable matrix

Square matrix A is said to diagonalizable if it is similar to diagonal matrix. i.e. $A = MDM^{-1}$ or $M^{-1}AM = D$.

+ Condition of existance of modal matrix M or condition of diagonalization

1) If all the ~~ideal~~ eigen values are distinct or different (A.M. = G.M = 1) then matrix is diagonalizable.

2) If eigen values are repeated then in that case if A.M = G.M = for each eigen value then matrix is diagonalizable.

+ Algebric Multiplicty (A.M)

The number of repetition of eigen value is called Algebric Multiplicty.

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* Geometric multiplicity
The number of linearly independant eigen vector of eigen value is called geometric multiplicity

Note :-

1) G.M is always positive and less than or equal to A.M

2) If eigen values are repeated then find eigen vector of ~~eig~~ repeated eigen values first.

3) Modal matrix and diagonal matrix will exist if matrix is digonalizable. (A.M. = G.M. for each eigen value) and it will of following form.

Ex.1) Check whether the following matrix $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{bmatrix}$ is diagonalizable or not. If the matrix is diagonalizable then find modal matrix and diagonal matrix.

→ Characteristic eq^n is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -5 & -2-\lambda \end{vmatrix} = 0$$

$$\lambda^3 + [4+3-2]\lambda^2 + [(4)(3) + (3)(-2) + (4)(-2) - (6)(1) - (6)(-1) - (2)(-5)]\lambda - 4 = 0$$

$$\lambda^3 + 5\lambda^2 + [12 - 6 - 8 - 6 + 6 + 10]\lambda - 4 = 0$$

$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$

Eigen values are $\lambda = 1, 2, 2$

A.M of $\lambda = 1$ is 1
A.M of $\lambda = 2$ is 2

Eigen vector for $\lambda = 2$

$$[A - \lambda I]X = 0$$
$$[A - 2I]X = 0$$

$$\begin{bmatrix} 2 & 6 & 6 \\ 1 & 1 & 2 \\ -1 & -5 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow[R_3 + \frac{1}{2}R_1]{R_2 - \frac{1}{2}R_1} \begin{bmatrix} 2 & 6 & 6 \\ 0 & -2 & -1 \\ 0 & -2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow{R_3 - R_2} \begin{bmatrix} 2 & 6 & 6 \\ 0 & -2 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2x_1 + 6x_2 + 6x_3 = 0$$
$$-2x_2 - x_3 = 0$$
$$-2x_2 = x_3$$

put $x_3 = t$

$$-2x_2 = t$$
$$x_2 = -\frac{t}{2}$$

$$2x_1 + 6\left(-\frac{t}{2}\right) + 6(t) = 0$$
$$2x_1 + 3t = 0$$
$$x_1 = -\frac{3t}{2}$$

Eigen vector is

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3t/2 \\ -t/2 \\ t \end{bmatrix}$$

$$= -t/2 \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$$

here,

G.M = 1 (we get only 1 independenly eigen vector)

& A.M = 2

$\therefore$ A.M $\neq$ G.M

matrix is non-diagonalizable.

Ex.2) Check whether matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$ is diagonalisable or not. If matrix is diagonalisable then find model matrix & diagonal matrix.

→

The characteristic eqn is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 2 \\ 3 & 3 & 4-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - [2+3+4]\lambda^2 + [6+12+8-2-3-6]\lambda - 7 = 0$$

$$\lambda^3 - 9\lambda^2 + 15\lambda - 7 = 0$$

$$\therefore \lambda = 7, 1, 1$$

$\therefore$ A.M of 7 is 1 & A.M of 1 is 2

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Eigen vector for $\lambda = 1$

$$[A - \lambda I]X = 0$$

$$\begin{bmatrix} 2-1 & 1 & 1 \\ 2 & 3-1 & 2 \\ 3 & 3 & 4-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + x_2 + x_3 = 0$$

$$\therefore x_1 = -x_2 - x_3$$

put $x_2 = t$ & $x_3 = s$

$$x_1 = -t - s$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -t-s \\ t \\ s \end{bmatrix} = \begin{bmatrix} -t \\ t \\ 0 \end{bmatrix} + \begin{bmatrix} -s \\ 0 \\ s \end{bmatrix}$$

$$= t \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

let $t = 1$ & $s = 1$

$$X = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore$ Both are linear independent

for eigen value $\lambda = 1$, A.M = G.M = 2

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Eigen vector for $\lambda = 7$

$$[A - \lambda I]X = 0$$

$$\begin{bmatrix} 2-7 & 1 & 1 \\ 2 & 3-7 & 2 \\ 3 & 3 & 4-7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -5 & 1 & 1 \\ 2 & -4 & 2 \\ 3 & 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-5x_1 + x_2 + x_3 = 0$$
$$2x_1 - 4x_2 + 2x_3 = 0$$

by Cramer's rule

$$\frac{x_1}{\begin{vmatrix} 1 & 1 \\ -4 & 2 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} -5 & 1 \\ 2 & 2 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} -5 & 1 \\ 2 & -4 \end{vmatrix}}$$

$$\frac{x_1}{6} = \frac{-x_2}{-12} = \frac{x_3}{18}$$

$$\frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{3}$$

$$X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

for eigen vector $\lambda = 7$, $A.M = G.M = 1$

A.M. = G.M for each eigen value. so matrix A is diagonalisable then

$$M = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix} \quad \& \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

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* Degenerate & non-degenerate sol^n.
If all the values obtained in basic feasible sol^n are non-zero then it is called non-degenerate sol^n, otherwise called degenerate sol^n. ~~Optimal~~ ~~so~~

* Optimal sol^n.
The basic feasible sol^n which optimizes which objective function, satisfy constraints and non-negetive restriction is called optimal sol^n.

Ex. 1] Consider the following problem Maximize $z = 2x_1 - 2x_2 + 4x_3 - 5x_4$ subject to $x_1 + 4x_2 - 2x_3 + 8x_4 = 2$
$-x_1 + 2x_2 + 3x_3 + 4x_4 = 1$. Determine all basic solutions ~~from a second~~ 2) all feasible basic
3) Optimal feasible basic sol^n.

→ no. of basic sol^n

no. of basic sol^n	non-basic variable $= 0$	basic variable	basic eq^n & values	is sol^n feasible	is sol^n degenerate	value of z	optimal sol^n
1	$x_3 = 0$ $x_4 = 0$	x_1 x_2	$x_1 + 4x_2 = 2$ $-x_1 + 2x_2 = 1$ $x_1 = 0,\ x_2 = \frac{1}{2}$	yes	yes	-1	
2	$x_2 = 0$ $x_4 = 0$	x_1 x_3	$x_1 - 2x_3 = 2$ $-x_1 + 3x_3 = 1$ $x_1 = 8,\ x_3 = 3$	yes	no	28	

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* Simplex Method.

1> Convert the given LPP in standard form of LPP.

Ex.] ~~Minimi~~ Solve the following LPP by simplex method
Minimize $z = x_1 - 3x_2 + x_3$ subject to $3x_1 - x_2 + 3x_3 \leq 7$, $2x_1 - 4x_2 \geq -12$, $-4x_1 + 3x_2 + 8x_3 \leq 10$,
$x_1, x_2, x_3 \geq 0$

→ Convert the given LPP in standard form as

$$\text{Max } z' = -z = -x_1 + 3x_2 - x_3$$
$$z' + x_1 - 3x_2 + x_3 = 0$$

$$\text{st} \quad 3x_1 - x_2 + 3x_3 + s_1 = 7$$
$$-2x_1 + 4x_2 + s_2 = 12$$
$$-4x_1 + 3x_2 + 8x_3 + s_3 = 10$$

check for smallest -ve no.

check for smallest +ve

iteration no.	basic variable	coefficient of x_1	x_2	x_3	s_1	s_2	s_3	r.h.s.	ration
0	z'	1	-3	1	0	0	0	0	
	s_1	3	-1	3	1	0	0	7	$7/-1 = -7$
s_2 leaves	s_2	-2	4* convert	0	0	1	0	12	$12/4 = 3$
x_2 enters	s_3	-4	3	8	0	0	1	10	$10/3 = 3.333$
1	z'	$-1/2$	0	1	0	$3/4$	0	9	
s_1 leaves	s_1	$5/2$ *	0	3	1	$1/4$	0	10	4
x_1 enters	x_2	$-1/2$	1	0	0	$1/4$	0	3	-6
	s_3	$-5/2$	0	8	0	$-3/4$	1	1	$-2/5$
2	z'	0	0	$8/5$	$1/5$	$4/5$	0	11	
	x_1	1	0	$6/5$	$2/5$	$1/10$	0	4	
	x_2	0	1	$3/5$	$1/5$	$3/10$	0	5	
	x_3	0	0	11	1	$-1/2$	1	11	

when -ve is removed from z then stop

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* NLPP with one equality constant (two or three) variables using method of ~~Lagrag~~ Lagranges multiplier

~~Pro~~ - Definition

An optimization problem in which either an objection function and or constraints (same or all) are non-linear is called Non-linear Programming Problem (NLPP)

- Procedure

Consider the NLPP as optimize $z = f(x_1, x_2)$

subject to $g(x_1, x_2) = b_1$

$x_1, x_2 \geq 0$

The procedure of solving above NLPP is as follows

① ~~write~~ express the constraints with r.h.s. = 0 as

~~$h(x_1, x_2)$~~ -

$$h(x_1, x_2) = g(x_1, x_2) - b = 0$$

② Construct lagrangian function using lagrangian multiplier λ as

$$L(x_1, x_2, h) = f(x_1, x_2) - \lambda h(x_1, x_2)$$

③ The maxima or minima denoted by X° can be obtained from following eqn

$$\frac{\partial L}{\partial x_1} = 0, \quad \frac{\partial L}{\partial x_2} = 0, \quad \frac{\partial L}{\partial \lambda} = 0$$

④ To determine weather obtained above is maxima or minima, we solve the following determinent.

$$\Delta_3 = \begin{vmatrix} 0 & \partial h/\partial x_1 & \partial h/\partial x_2 \\ \partial h/\partial x_1 & \partial^2 L/\partial x_1^2 & \partial^2 L/\partial x_1 \partial x_2 \\ \partial h/\partial x_2 & \partial^2 L/\partial x_2 \partial x_1 & \partial^2 L/\partial x_2^2 \end{vmatrix}$$

If Δ_3 is ~~de~~ positive at X_0^* then X_0^* is maxima and if Δ_3 is negative then X_0^* is minima.

Ex. 1] Using the method of Lagrange~~s~~ multiplier, solve the following NLPP.

optimise $z = 6x_1^2 + 5x_2^2$

subject to $x_1 + 5x_2 = 7$

$x_1, x_2 \geq 0$

→

① $h(x_1, x_2) = x_1 + 5x_2 - 7 = 0$

② construct ~~langra~~ lagrangian fuction as

$L(x_1, x_2, \lambda) = f(x_1, x_2) - \lambda h(x_1, x_2)$

$L(x_1, x_2, \lambda) = 6x_1^2 + 5x_2^2 - \lambda(x_1 + 5x_2 - 7)$

③ The point of maxima or minima can be obtained by solving following eq^ns.

$\frac{\partial L}{\partial x_1} = 0$	$\frac{\partial L}{\partial x_2} = 0$	$\frac{\partial L}{\partial \lambda}$
$12x_1 - \lambda = 0$	$10x_2 - 5\lambda = 0$	$-(x_1 + 5x_2 - 7) = 0$
$12x_1 = \lambda$	$2x_2 = \lambda$	$x_1 + 5x_2 - 7 = 0$
$12x_1 - 2x_2 = 0$		$x_1 + 5x_2 = 7$
$6x_1 = x_2$	$\boxed{x = \frac{84}{31}}$	$x_1 + 5 \times 6x_1 = 7$

$x_1 + 30x_1 = 7$

$31x_1 = 7$

$x_1 = 7/31$ — ①

$5x_2 = 7 - 7/31$

$x_2 = \frac{42}{31}$ — ②

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The point of maxima or minima i.e. X_0 is $X_0\left(\frac{7}{31}, \frac{42}{31}\right)$

$$\Delta_3 = \begin{vmatrix} 0 & \partial h/\partial x_1 & \partial h/\partial x_2 \\ \frac{\partial h}{\partial x_1} & \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} \\ \frac{\partial h}{\partial x_2} & \frac{\partial^2 L}{\partial x_2 \partial x_1} & \frac{\partial^2 L}{\partial x_2^2} \end{vmatrix}$$

$$\frac{\partial h}{\partial x_1} = 1 \qquad \frac{\partial h}{\partial x_2} = 5$$

$$\frac{\partial L}{\partial x_1} = 12x_1 - \lambda \qquad \frac{\partial^2 L}{\partial x_1^2} = 12$$

$$\frac{\partial^2 L}{\partial x_2 \partial x_1} = \frac{\partial}{\partial x_2}\left(\frac{\partial L}{\partial x_1}\right) = 0$$

$$\frac{\partial L}{\partial x_2} = 10x_2 - 5\lambda \qquad \frac{\partial^2 L}{\partial x_2^2} = 10$$

$$\Delta_3 = \begin{vmatrix} 0 & 1 & 5 \\ 1 & 12 & 0 \\ 5 & 0 & 10 \end{vmatrix}$$

$$\Delta_3 = -310$$

Δ_3 is negative so we get,

$$Z_{min} = \frac{294}{31}, \quad x_1 = \frac{7}{31}, \quad x_2 = \frac{42}{31}$$

* NLPP with 1 constraint and 3 variables using method of Lagranges multiplier.

The Maxima and minima (z_{max} & z_{min}) depends on determinent Δ_4

$$\Delta_4 = \begin{vmatrix} 0 & \frac{\partial h}{\partial x_1} & \frac{\partial h}{\partial x_2} & \frac{\partial h}{\partial x_3} \\ \frac{\partial h}{\partial x_1} & \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} & \frac{\partial^2 L}{\partial x_1 \partial x_3} \\ \frac{\partial h}{\partial x_2} & \frac{\partial^2 L}{\partial x_2 \partial x_1} & \frac{\partial^2 L}{\partial x_2^2} & \frac{\partial^2 L}{\partial x_2 \partial x_3} \\ \frac{\partial h}{\partial x_3} & \frac{\partial^2 L}{\partial x_3 \partial x_1} & \frac{\partial^2 L}{\partial x_3 \partial x_2} & \frac{\partial^2 L}{\partial x_3^2} \end{vmatrix}$$

- If Δ_3 and Δ_4 are both negative then X_0 is minima
- If Δ_3 is positive and Δ_4 is negative then X_0 is maxima.

Ex. 1) Using method of Lagranges multiplier solve the following NLPP :

optimize $z = x_1^2 + x_2^2 + x_3^2 - 10x_1 - 6x_2 - 4x_3$

subject to $x_1 + x_2 + x_3 = 7$

$x_1, x_2, x_3 \geq 0$

$\rightarrow$ $h(x_1, x_2, x_3) = x_1 + x_2 + x_3 - 7 = 0$

Construct lagrangian fun as

$L(x_1, x_2, x_3, \lambda) = f(x_1, x_2, x_3) - \lambda h(x_1, x_2, x_3)$

$L = x_1^2 + x_2^2 + x_3^2 - 10x_1 - 6x_2 - 4x_3 - \lambda[x_1 + x_2 + x_3 - 7]$

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Maxima or minima can be obtained as

$\frac{\partial L}{\partial x_1} = 0$	$\frac{\partial L}{\partial x_2} = 0$	$\frac{\partial L}{\partial x_3} = 0$	$\frac{\partial L}{\partial \lambda} = 0$
$2x_1 - 10 - \lambda = 0$ —①	$2x_2 - 6 - \lambda = 0$ —②	$2x_3 - 4 - \lambda = 0$ —③	$-(x_1 + x_2 + x_3 - 7) = 0$ $x_1 + x_2 + x_3 = 7$ —④

add ①, ②, ③ & use eq^n ④

$2[x_1 + x_2 + x_3] - 20 - 3\lambda = 0$

$2 \times 7 - 20 - 3\lambda = 0$

$-6 - 3\lambda = 0$

$\lambda = -2$

put $\lambda = -2$ in ①, ② & ③

$2x_1 - 8 = 0$ $x_1 = 4$	$2x_2 - 4 = 0$ $x_2 = 2$	$2x_3 - 2 = 0$ $x_3 = 1$

The point of maxima or minima i.e. X_0 is
(

$$\Delta_4 = \begin{vmatrix} 0 & \frac{\partial h}{\partial x_1} & \frac{\partial h}{\partial x_2} & \frac{\partial h}{\partial x_3} \\ \frac{\partial h}{\partial x_1} & \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} & \frac{\partial^2 L}{\partial x_1 \partial x_3} \\ \frac{\partial h}{\partial x_2} & \frac{\partial^2 L}{\partial x_2 \partial x_1} & \frac{\partial^2 L}{\partial x_2^2} & \frac{\partial^2 L}{\partial x_2 \partial x_3} \\ \frac{\partial h}{\partial x_3} & \frac{\partial^2 L}{\partial x_3 \partial x_1} & \frac{\partial^2 L}{\partial x_3 \partial x_2} & \frac{\partial^2 L}{\partial x_3^2} \end{vmatrix}$$

$$x_1 + 2x_1 - \frac{2x_1}{3} = 20 - 1 + \frac{8}{3} + \frac{4x_1 + 10}{3}$$

$$x_1 + 2x_1 - \frac{2x_1}{3} - \frac{4x_1}{3} = 20 - 1 + \frac{8}{3} + \frac{10}{3}$$

$x_1 =$

$$\Delta_4 = \begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{vmatrix}$$

$$= 0 - 1\begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} + 1\begin{vmatrix} 1 & 2 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 2 \end{vmatrix} - 1\begin{vmatrix} 1 & 2 & 0 \\ 1 & 0 & 2 \\ 1 & 0 & 0 \end{vmatrix}$$

$$= -4 - 4 - 4$$

$$\Delta_4 = -12$$

$$\Delta_3 = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix}$$

$$\Delta_3 = -4$$

$\Delta_4 = -12$ & $\Delta_3 = -4$ both are negative so we get point of minima $X_0 (4, 2, 1)$

$$Z_{min} = 4^2 + 2^2 + 1^2$$

$$=$$

& $x_1 = 4$, $x_2 = 2$, $x_3 = 1$

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Ex.] Find the fourier series of $f(x) = 4 - x^2$ in the interval $(0, 2)$.

$\rightarrow$ $f(x) = a_0 + \sum a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1} b_n \cos\left(\frac{n\pi x}{l}\right)$

where $a_0 = \frac{1}{2l} \int_c^{c+2l} f(x)\, dx$, a_n, b_n

Comparing the given interval $(0, 2)$ with $(c, c+2l)$

$c = 0$ | $c + 2l = 2$
$0 + 2l = 2$
$\boxed{l = 1}$

The fourier coefficient can be written as

$$a_0 = \frac{1}{2l} \int_c^{c+2l} f(x)\, dx = \frac{1}{2(1)} \int_0^2 (4 - x^2)\, dx$$

$$= \frac{1}{2}\left[4x - \frac{x^3}{3}\right]_0^2$$

$$= \frac{1}{2}\left[(8 - 0) - \left(\frac{2^3}{3} - 0\right)\right] \quad \text{or} \quad \frac{1}{2}\left[\left(8 - \frac{2^3}{3}\right) - (0 - 0)\right]$$

$$= \frac{1}{2}\left[8 - \frac{8}{3}\right] = \frac{8}{2}\left[1 - \frac{1}{3}\right]$$

$$= \frac{8}{2}\left[\frac{2}{3}\right]$$

$$\boxed{a_0 = \frac{8}{3}}$$

$\sin 2n\pi = 0$
$\cos 2n\pi = 1$

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$$\int uv = (u)(v_1) - (u')(v_2) + (u'')(v_3) - \dots$$

$$a_n = \frac{1}{l}\int_{c}^{c+2l} f(x)\cos\left(\frac{n\pi x}{l}\right)dx$$

$$= \frac{1}{1}\int_{0}^{2}(4-x^2)\cos\left(\frac{n\pi x}{1}\right)dx$$

$$= \int_{0}^{2}(4-x^2)\cos(n\pi x)\,dx$$

$$= \left[(4-x^2)\left(\frac{\sin(n\pi x)}{n\pi}\right) - (-2x)\left(-\frac{\cos(n\pi x)}{n^2\pi^2}\right) + (-2)\left(\frac{-\sin(n\pi x)}{n^3\pi^3}\right)\right]_0^2$$

$$= \left[\left(0 - \frac{4}{n^2\pi^2} + 0\right) - (0 - 0 + 0)\right]$$

$$a_n = \frac{-4}{n^2\pi^2}$$